Quotation: TEST INPUT - readiness trace pump panel

Job number: TEST-001

Boards: PP1

Currency: JOD

Description	Qty	Unit	Unit price	Total
Circuit-breaker 4P C100 -Q1	1	pcs	not priced	-
Circuit-breaker 3P C16 -Q2	1	pcs	not priced	-
Residual current circuit-breaker 4P 16A 30mA -F1	1	pcs	not priced	-
Circuit-breaker 1P C10 -Q3	1	pcs	not priced	-
ABB T2S160 MA 20 -Q4	1	pcs	not priced	-
ABB A30 -Q5 -Q9	2	pcs	not priced	-
ABB TA25DU19 -F2	1	pcs	not priced	-
ABB T2S160 MA80 -Q6	1	pcs	not priced	-
ABB A63 -Q7 -Q8	2	pcs	not priced	-
ABB TA75DU42 -F3	1	pcs	not priced	-
Bussmann 170M1311 -F4	1	pcs	not priced	-
ABB ACS880-01-12A6-3 -T1	1	pcs	not priced	-
Cable 3G1.5 Cu PVC -W1	10	m	not priced	-
Cable 4G1.5 Cu PVC -W2 -W4	50	m	not priced	-
Cable 7G6 Cu PVC -W3	30	m	not priced	-

Incomplete: these lines have no price and are not in the total:

circuit_breaker:4P:C100 (no price)

circuit_breaker:3P:C16 (no price)

residual_current_device:4P:16A:30mA (no price)

circuit_breaker:1P:C10 (no price)

T2S160 MA 20 (no price)

A30 (no price)

TA25DU19 (no price)

T2S160 MA80 (no price)

A63 (no price)

TA75DU42 (no price)

170M1311 (no price)

ACS880-01-12A6-3 (no price)

cable:3G1.5:Cu:PVC (no price)

cable:4G1.5:Cu:PVC (no price)

cable:7G6:Cu:PVC (no price)

Materials	0.00
Labour and enclosure	0.00
Markup	0.00
VAT	0.00
Total (JOD)	**0.00**

Findings (0 blocking, 9 warning, 10 info)

[WARNING] PP1 / c2: Duty pump (test): Ir 15.2 A, the typical Ir of a 7.5 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate. (motor_current_table; source:

policy; engineering confirmation required)

[WARNING] PP1 / c3: Standby pump (test): Ir 56 A, the typical Ir of a 30 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate. (motor_current_table; source:

policy; engineering confirmation required)

[WARNING] PP1 / c4: Booster (test): Ir 10.38 A from 5.5 kW shaft power at cos phi 0.85, efficiency 0.9; check it against the nameplate. (motor_current_formula; source: policy; engineering confirmation required)

[WARNING] PP1 / c4: Booster (test): the fuses need at least 75 A prospective short-circuit current at the board to operate fast enough. (drive_fuse_fault_level; source: policy; engineering confirmation required)

[WARNING] PP1: Loads without a power factor were taken at cos phi 0.9. (default_power_factor; source: policy; engineering confirmation required)

[WARNING] PP1: No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze. (earth_fault_no_ze; source: policy;

engineering confirmation required)

[WARNING] PP1: Each cable's withstand k^2S^2 is in the cable list; compare it with the selected breaker's let-through I^2t at 10 kA from its maker's curve (ABB handbook §2.4). (short_circuit_withstand; source: policy;

engineering confirmation required)

[WARNING] PP1: Circuit rules are this software's defaults, not confirmed by the company's engineers. (rules_unconfirmed; source: policy; engineering confirmation required)

[WARNING] PP1: Enclosure: none named, since 11 devices have no width (rail_widths_mm in the company settings). (enclosure_unknown_widths; source: policy; engineering confirmation required)

[INFO] PP1 / c4: Booster (test): the cable after a drive is protected by the drive's motor overload setting, which is not checked here. (integrity_drive_cable_unchecked; source: policy)

[INFO] PP1: Group 1 breaker raised to 16 A, 1.6 x the 10 A breaker after it, for discrimination. (discrimination_group_raised; source: policy)

[INFO] PP1 / c2: Duty pump (test): dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2. (starter_table; source: policy)

[INFO] PP1 / c3: Standby pump (test): star_delta starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 5: 400 V 50 kA Y/D Normal Type 2. (starter_table; source: policy)

[INFO] PP1 / c3: Standby pump (test): star and delta contactors need a mechanical interlock; the motor is fed by six conductors (U1 V1 W1, U2 V2 W2). (star_delta_interlock; source: policy)

[INFO] PP1 / c4: Booster (test): ACS880-01-12A6-3 for normal duty at 30 °C, behind 25 A aR fuses (ABB ACS880-01 hardware manual (3AUA0000078093)). Use a screened, symmetrical motor cable, earthed 360° at both ends.

(drive_selected; source: policy)

[INFO] PP1: Breakers are rated for at least 10 kA breaking capacity against 10 kA prospective at the board; each RCCB's conditional short-circuit current with the breaker ahead of it must reach it too.

(breaking_capacity; source: policy)

[INFO] PP1: Leave 1 spare outgoing ways (20 %). (spare_ways; source: policy)

[INFO] PP1: Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip

(about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables. (discrimination; source: policy)

[INFO] PP1: Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation. (cable_conditions; source: policy)